

Beyond Accuracy: Comprehensive Mathematical Modeling for Health Insurance Pricing Recommendation, Guided by System Identification and Benchmarked Against an AI-Generated Workflow

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Abstract

This paper presents an applied mathematical modeling study for health insurance pricing recommendation. Using a public Kaggle dataset with 1,338 anonymized real customer records from a health plan operator, the study separates two decisions that are frequently conflated in pricing projects: setting the initial premium for new customers and readjusting premiums for current customers. K-Means, Linear Regression, candidate families of nonlinear static models guided by System Identification principles, Kernel Ridge, and a Random Forest produced by an AI workflow were compared. On the 268-customer holdout, Random Forest achieved the best predictive accuracy (RMSE = R\$ 647.89; $R^2 = 0.797$), while Kernel Ridge was the strongest alternative selected by the formal protocol before the AI benchmark (RMSE = R\$ 753.18; $R^2 = 0.726$). For current customers, it is shown that an expense-prediction model should not be applied directly as a readjustment policy: the problem becomes one of controlling churn and preserving margin. The central contribution is a framework connecting modeling, validation, pricing and churn, showing that higher accuracy does not replace governance and economic judgement.

Keywords: health insurance; pricing; System Identification; machine learning; churn; Random Forest; Kernel Ridge.

1 Introduction

Health insurance pricing is simultaneously a statistical, an economic and a managerial problem. The company must estimate the expected cost of each customer, set a premium compatible with the desired margin, and avoid excessive readjustments that trigger cancellation. In data projects, these three decisions are usually compressed into a single question: which model best predicts medical expense? This question is necessary, but insufficient. A model with the lowest average error may produce price recommendations that raise churn, reduce profit, or weaken regulatory justification.

This work proposes a more comprehensive formulation. First, the arrival of new customers is treated as an expense-prediction problem. Second, the existing portfolio is treated as a readjustment problem under a churn constraint, since for existing customers both the current premium and the realized expense are already observed. Third, a modeling approach guided by System Identification principles is compared with an AI-generated workflow, bringing the study closer to a real situation in which a manager might ask an AI system to automatically build a prediction model.

The contribution of the article is to show that the discussion must not end at RMSE, MAE or R^2 . Such metrics indicate predictive quality, but a price recommendation also requires a function that transforms the prediction into a premium, a churn hypothesis, a policy for loss-generating customers,

and an auditability analysis. The study therefore organizes the problem as a decision chain: data, structural choice, validation, price, churn and profit.

2 Related work and background

2.1 Pricing with explainable machine learning

Recent work in health insurance shows growing interest in explainable machine learning, that is, predictive models accompanied by techniques that help interpret the determinants of cost or premium. Orji and Ukwandu (2024) compare Random Forest, Gradient Boosting and XGBoost for medical-cost prediction, using SHAP and ICE to interpret relevant variables. Kapse et al. (2025) also address premium customization with machine learning and explainable AI. At industrial scale, Kshirsagar et al. (2021) report interpretable models for transparent health-plan pricing, with performance superior to the insurer's internal actuarial model.

The present work dialogues with this literature but emphasizes one difference: expense prediction is treated as an intermediate step of a price recommendation. Therefore, beyond accuracy and interpretability, the effects on churn and profit are evaluated, distinguishing new customers from current customers.

2.2 System Identification as a modeling guide

System Identification, as presented by Ljung (1999) and Aguirre (2014), offers a formal discipline for choosing the structure of a model from the behavior of the data, instead of selecting algorithms informally. It is important to clarify that this literature is not restricted to dynamic systems: although the field grew within control engineering, Aguirre (2014) explicitly treats nonlinear static models — polynomial, rational, spline, neural-network, fuzzy and kernel-based —, and the discipline's decision tree branches, at the very start, between the dynamic and the static case. This study relies precisely on the static branch of that tree, which is the formally appropriate part of the framework for the problem analyzed here.

The model-selection methodology of Ljung and Aguirre can be summarized in four steps. First, classify the system as dynamic or static, observing whether there is temporal dependence among observations. Second, test whether a linear structure is sufficient, for example by comparing the R^2 of a linear model with that of a degree-two polynomial. Third, respect the parsimony principle, preferring the simplest model compatible with the data — Ljung (1999) suggests, as a rule of thumb, at least five observations per estimated parameter. Fourth, validate performance out of sample, on data not used for estimation. The final criterion is not merely the smallest fitting error, but the combination of data fit, structural simplicity and generalization ability.

This work applied exactly this sequence. The dataset analyzed is cross-sectional — each row represents a customer at one instant, with no explicit temporal memory —, so the problem is static by construction. Next, the sufficiency of linear relations was tested, candidate families of static models were compared, and performance was validated on the 268-customer holdout. Kernel Ridge is not presented as a direct recommendation of Ljung or Aguirre; it enters as a coherent candidate within the nonlinear static-model step, by representing a flexible regression surface via an RBF kernel and L2 regularization.

2.3 Generative AI and data-science workflows

The use of language models to automate data-analysis pipelines has become an active research front. Gu et al. (2024) review the use of LLMs in machine-learning workflows, including feature engineering, model selection, hyperparameter optimization and evaluation. Jansen et al. (2025) show that LLMs can generate analysis pipelines, but also that task complexity reduces the reliability of executable code. Zeng et al. (2025) stress that, in data science, observing the result is not enough; one must assess the reproducibility and the logical plan of the workflow.

In this study, the AI-generated workflow is treated as an external benchmark. It represents the answer a manager without statistical training could obtain by asking, in natural language, for a method to predict the expenses of future customers. This comparison is relevant because it tests not only accuracy, but also what the AI does not deliver automatically: statistical diagnostics, separation of business problems, governance and a policy for current customers.

2.4 Premium readjustment and churn

The relationship between premium increases and cancellation is central to private health insurance. Souza et al. (2025) study, by regression discontinuity, the effect of age-band readjustments on cancellation and plan switching in Brazil, showing that price is a relevant determinant of beneficiary exit. This result justifies incorporating churn into the price recommendation. In the present project, the churn function implemented in the notebook is an operational parameterization inspired by that discontinuity logic; therefore, the results should be interpreted as a decision simulation, not as original causal estimation.

3 Materials and methods

3.1 Data

The dataset is public, obtained from Kaggle, and contains 1,338 anonymized real customer records from a health plan operator. The operator's identity is not disclosed and must remain confidential. The available variables are age, gender, body mass index, number of children, discount eligibility, region, medical expenses and premium charged.

Variable	Meaning	Type	Use in the study
age	customer age	numeric	risk profile and expense prediction
gender	gender	categorical	encoding for supervised models
bmi	body mass index	numeric	health risk
children	number of children	discrete	family structure and expected usage
discount_eligibility	discount eligibility	binary	segmentation and profile composition
region	region (location)	categorical	regional control
expenses	medical expenses	numeric	prediction target
premium	premium (monthly fee charged)	numeric	readjustment base for current customers

Table 1 - Variables used in the study.

Throughout the text, the term premium denotes the monthly fee charged to the customer; the two terms are used as synonyms. The dataset was split randomly into two parts, with `random_state = 1` to ensure the reproducibility of the experiment: 80% of the records (1,070 customers) formed the training set, used to estimate the models, and the remaining 20% (268 customers) made up the test set, kept

completely separate throughout the fitting. It is this 20% set that is referred to, throughout the article, as the 268-customer holdout: it is used only in the final evaluation, so that every reported metric measures out-of-sample performance. Figure 1 shows that the expense variable has a right tail, with positive skewness of 0.749. This is important because simple linear models can be affected by high-expense observations and by unspecified nonlinear relations.

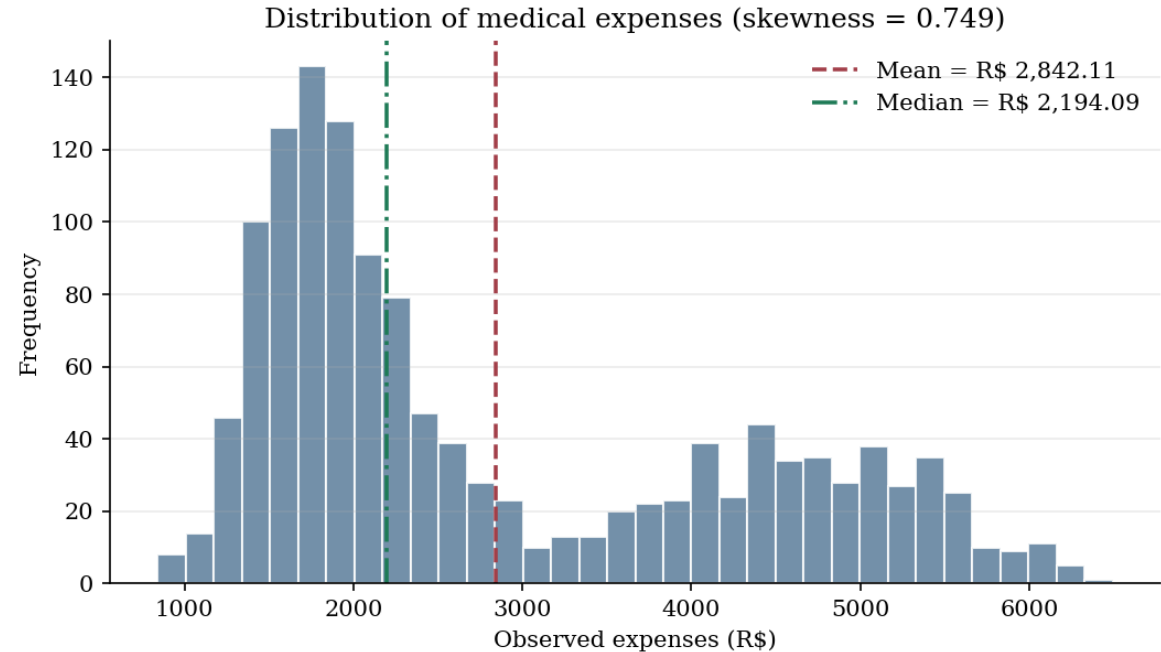


Figure 1 - Distribution of observed medical expenses in the public dataset.

3.2 Models evaluated

Segmentation, regression and machine-learning approaches were evaluated. K-Means was used as a segmentation baseline, assigning each customer the cluster's mean expense. Linear Regression served as an interpretable baseline. Candidate families of linear and nonlinear static models were then tested: degree-two Polynomial Regression, Ridge, Lasso, Natural Spline, MLP and Kernel Ridge. Finally, a Random Forest produced by an AI workflow was compared as an external benchmark; this workflow was produced by vibe coding in the Claude Code tool (Anthropic), with the Opus 4.7 model, and is detailed in Section 4.5. For a fair comparison, the AI workflow received exactly the same training set (the 80% of customers) used by the other models and was evaluated on the same 268-customer holdout (the 20% test set).

Family	Role in the study	Methodological note
K-Means (Clustering)	segmentation	does not optimize regression error; predicts the group mean
Linear Regression	explainable baseline	additive and linear model
Polynomial degree 2	curvature test	tested whether quadratic terms added value
Ridge/Lasso	regularization	instability control and term selection
Natural Spline	smooth nonlinear	flexibility with interpretable structure
MLP	flexible nonlinear	feedforward neural network
Kernel Ridge	regularized nonlinear	RBF kernel with L2 penalty
Random Forest (AI)	external benchmark	selected by an AI-generated workflow

Table 2 - Families of models evaluated.

3.3 Mathematical formulation

In Linear Regression, expenses are approximated by an additive combination of the explanatory variables:

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i$$

In K-Means, each customer is assigned as prediction the mean expense of the cluster to which it belongs:

$$\hat{y}_i = \bar{y}_{c_i}$$

In Kernel Ridge with an RBF kernel, the prediction is a weighted sum of similarities between the customer analyzed and the training points:

$$\hat{y}(x) = \sum_i \alpha_i \exp(-\gamma \|x - x_i\|^2), \quad \alpha = (K + \lambda I)^{-1} y$$

In Random Forest, the regression prediction is the average of the predictions produced by the B trees of the ensemble:

$$\hat{y}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

The transformation of estimated expense into price used a desired margin of 25%:

$$P_i = \frac{\hat{y}_i}{1-m}, \quad m = 0.25$$

The operational churn function used in the notebook considers the percentage increase of the premium relative to the current premium:

$$c(\Delta_i) = \alpha + \beta_1 x_i + \beta_2 I(x_i \geq 0.20) + \beta_3 (x_i - 0.20) I(x_i \geq 0.20)$$

where $x_i = \Delta_i/100$, $\alpha = 0.05$, $\beta_1 = 1$, $\beta_2 = 0.15$ and $\beta_3 = 1.5$. The function is truncated to the interval $[0, 1]$.

Finally, writing Linear Regression with the specific variables of the dataset, the estimated structure for each customer's expense takes the form:

$$\text{expenses} = \beta_0 + \beta_1 \text{age} + \beta_2 \text{bmi} + \beta_3 \text{children} + \beta_4 \text{gender} + \beta_5 \text{discount} + \beta_6 \text{region} + \varepsilon$$

where age, bmi and children enter directly as numeric variables, while gender, discount (discount eligibility) and region are categorical variables encoded as binary (dummy) variables; region, having four categories, expands into three binary variables. The term ε represents the random error. The other nonlinear families operate on the same vector of variables, changing only the functional form that relates those variables to expenses.

4 Results

4.1 Target variable and per-customer predictions

Before the formal selection, it is worth clarifying what each model actually estimated. In every case — K-Means, Linear Regression, Kernel Ridge and Random Forest — the target variable is the same: each customer's medical expense (expenses). Table 3 illustrates, for a sample of holdout customers, the true expense next to the expense predicted by each model. Only a few rows are shown; the metrics reported in the following sections are computed over all 268 holdout customers.

Customer	True expense (R\$)	K-Means (R\$)	Lin. Regression (R\$)	Kernel Ridge (R\$)	Random Forest (R\$)
1	1,653.18	1,893.97	1,855.39	1,835.42	1,900.37
2	5,589.00	4,713.26	4,549.25	4,859.94	4,634.04
3	5,159.52	4,713.26	4,495.62	4,979.05	4,828.12
4	3,315.35	2,097.74	3,035.32	2,516.18	2,281.50
5	1,413.54	1,893.97	1,559.28	1,711.26	1,834.92
6	2,569.80	2,618.75	1,860.84	1,746.13	2,039.84
7	2,393.93	2,097.74	3,524.56	3,145.19	2,220.81
8	5,548.98	4,713.26	4,333.43	4,708.59	4,486.73

Table 3 - True expense and expense estimated by each model for a sample of 8 of the 268 holdout customers.

The per-customer reading of Table 3 shows that, for most customers, the four models reach close values to one another — this is the case of Customer 1, whose four estimates cluster in a narrow band around the true expense. Divergence appears in the atypical-expense cases. An atypical expense is understood here as that of a customer whose observed spending is hard to explain from their risk profile — age, body mass index and number of children: for example, a young, apparently low-risk customer who nevertheless records a high expense, or the opposite. At these points each model reacts differently — K-Means assigns the cluster mean, whereas flexible models such as Kernel Ridge and Random Forest fit the prediction surface locally — and the estimates drift apart from one another.

This divergence is visible in Table 3 itself. For Customer 4, the true expense is R\$ 3,315.35, yet the four model estimates spread between R\$ 2,097.74 and R\$ 3,035.32; for Customer 7, the true expense is R\$ 2,393.93 and the estimates range from R\$ 2,097.74 to R\$ 3,524.56 — a difference of more than R\$ 1,426.82 between the lowest and the highest predicted value. These two customers contrast with Customer 1, where the highest and the lowest estimate differ by only a few tens of reais. It is precisely in these atypical cases that the choice of model starts to matter, because that is where the individual errors concentrate. This individual view makes Table 4, which summarizes each model's performance over the whole holdout set, more transparent.

4.2 Formal model selection

In the internal selection step, Kernel Ridge achieved the best performance among the candidate families evaluated before the AI benchmark. Table 4 presents the results on the internal split of the training set, with an 80/20 validation and cross-validation where applicable.

Model	R ² train	R ² val.	RMSE val.	MAE val.	CV R ²
Kernel Ridge	0.76	0.78	R\$ 650.53	R\$ 498.95	0.7161 +/- 0.0472

Model	R ² train	R ² val.	RMSE val.	MAE val.	CV R ²
Natural Spline	0.70	0.73	R\$ 712.24	R\$ 563.41	0.6869 +/- 0.0334
MLP	0.68	0.71	R\$ 742.52	R\$ 550.83	0.6194 +/- 0.0521
Polynomial degree 2	0.69	0.69	R\$ 761.45	R\$ 580.64	-
Linear Regression	0.61	0.68	R\$ 776.10	R\$ 604.86	0.6020 +/- 0.0455
Ridge	0.61	0.68	R\$ 777.15	R\$ 604.96	0.6025 +/- 0.0466
Lasso	0.61	0.68	R\$ 778.35	R\$ 604.96	0.6037 +/- 0.0471

Table 4 - Internal selection of candidate static models.

The result indicates that polynomial terms were not enough to surpass Kernel Ridge and that regularized linear models remained close to the linear baseline. Thus, the formal selection favored a regularized nonlinear structure, consistent with the hypothesis that local interactions and smoothness in the feature space are relevant for medical expenses.

4.3 Performance on the 268-customer holdout

After selection, Kernel Ridge was re-estimated on the full training set and evaluated on the 268-customer holdout. The Random Forest produced by the AI workflow was then evaluated on the same set. Table 5 shows the final metrics, without mixing average rows with model rows.

Model	MAE	RMSE	R ²	MAPE	Mean bias
K-Means (Clustering)	R\$ 617.08	R\$ 803.17	0.689	26.19%	10.42%
Linear Regression	R\$ 695.91	R\$ 893.20	0.615	30.13%	12.51%
Kernel Ridge	R\$ 572.83	R\$ 753.18	0.726	23.86%	7.90%
Random Forest (AI)	R\$ 496.95	R\$ 647.89	0.797	20.77%	6.86%

Table 5 - Predictive metrics on the common 268-customer holdout.

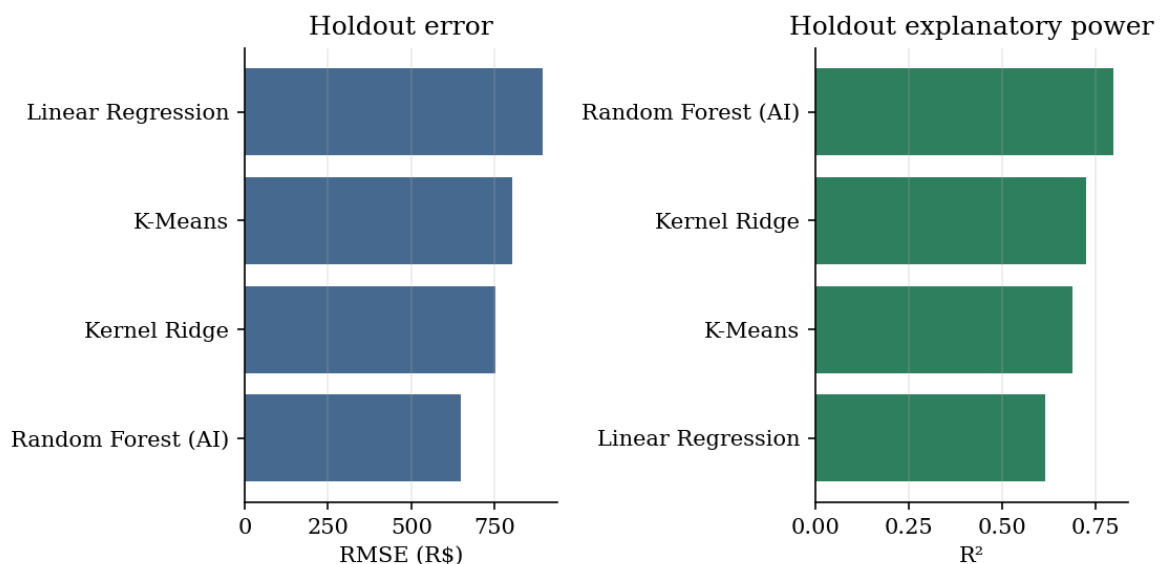


Figure 2 - Comparison of RMSE and R² on the holdout.

Random Forest had the lowest RMSE and the highest R² on the holdout. Therefore, for new customers, it is the best predictor among the models compared. Kernel Ridge, in turn, kept better performance than K-Means and Linear Regression, and has a more direct mathematical formulation for

auditing than a forest of trees.

4.4 Counterintuitive result: K-Means above Linear Regression

Linear Regression had a higher RMSE than K-Means on the holdout. This result does not imply that clustering is, in general, better than supervised regression for prediction. The most plausible interpretation is that plain OLS was misspecified for this dataset: there is a right tail in expenses, nonlinear effects and unspecified interactions. K-Means, although coarse, groups customers by profile proximity and may partially capture combinations of age, BMI and children that the linear model forces into a single average coefficient.

This finding is scientifically useful because it shows the risk of using a linear baseline merely for interpretability. When the target distribution is skewed and the response varies locally in the feature space, segmentation can outperform a misspecified linear regression, even though it remains inferior to nonlinear models fitted directly for predictive loss.

4.5 AI-generated workflow

The term *vibe coding*, coined by Karpathy in 2025, describes the paradigm in which the developer states the goal in natural language and a language model writes all the code — including the choice of algorithm and the tuning of hyperparameters — with little or no human technical review. The AI benchmark of this study followed exactly that procedure: the workflow was generated by *vibe coding* in the Claude Code tool, by Anthropic, using the Opus 4.7 model. A method to predict the expenses of future customers was requested in natural language, without guiding the statistical approach and without reviewing the generated code.

The prompt sent to Claude Code reproduces the speech of a manager with no statistical training and was deliberately kept free of any technical guidance (it was issued in Portuguese; an English translation is shown below):

I am the manager of a health insurance company. I am passing you the file "clientes_atuais.csv" where age = customer age; gender = customer sex; bmi = customer BMI; children = number of children of the customer; region = where the customer lives; expenses = the customer's monthly spending; premium = how much we currently charge. I want to know what amount to charge the next customers (premium). To do so, first I need to predict what the likely value of "expenses" of the new customers will be. How do I compute the likely value of "expenses" of the new customers given the values ("age, gender, bmi, children, region") I obtain from the new customers? I want the best possible prediction value.

The dataset provided to Claude Code was exactly the same one used by the other models: the 80% of customers in the training set described in Section 3.1. No information from the test set was exposed to the tool while the model was being built. Once the workflow was complete, the resulting Random Forest was applied to the same 20% of customers in the test set — the 268-customer holdout — so that its effectiveness could be compared, on equal terms, with that of the models selected by the formal protocol, as already presented in Sections 4.1 to 4.3.

The AI workflow compared five algorithms with cross-validation and GridSearchCV tuning, selecting a Random Forest with 300 trees, maximum depth 8, feature fraction 0.5 and a minimum of 5 samples per split. The workflow itself estimated an internal RMSE of R\$ 546.19; on the real holdout, RMSE was R\$ 647.89. The difference represents internal-validation optimism, but does not negate the main result: on the common holdout, Random Forest won on accuracy.

The implication is twofold. The AI was competent at finding a strong model, consistent with the literature that highlights tree ensembles as robust candidates in many tabular problems. However, it did

not automatically deliver a discussion of residual diagnostics, the separation between new and current customers, churn, governance or a readjustment policy.

4.6 Churn when applying model-derived prices

The expense estimated by each model is converted into a price by the 25%-margin formula presented in Section 3.3. Table 6 details this step for a sample of customers: for each customer, it shows the true expense, the premium currently charged and the premium suggested by each model — obtained by dividing the estimated expense (presented in Table 3) by the factor $(1 - 0.25)$. It is worth noting that the comparison with K-Means is fair: although clustering adopts cluster-specific target margins, the average of those margins equals approximately 25%, aligning it with the other models.

Customer	True expense	Current premium	K-Means	Lin. Regression	Kernel Ridge	Random Forest
1	1,653.18	1,944.92	2,525.30	2,473.85	2,447.23	2,533.83
2	5,589.00	5,322.85	6,284.35	6,065.66	6,479.92	6,178.72
3	5,159.52	4,913.83	6,284.35	5,994.16	6,638.73	6,437.49
4	3,315.35	4,420.46	2,996.78	4,047.10	3,354.91	3,042.00
5	1,413.54	1,662.99	2,525.30	2,079.04	2,281.68	2,446.56
6	2,569.80	3,426.41	3,273.44	2,481.12	2,328.17	2,719.79
7	2,393.93	3,191.90	2,996.78	4,699.42	4,193.59	2,961.08
8	5,548.98	5,284.74	6,284.35	5,777.91	6,278.12	5,982.31

Table 6 - True expense, current premium and premium suggested by each model (25% margin), in R\$, for a sample of 8 of the 268 holdout customers.

The prices thus derived are submitted to the churn function to classify each customer as retained or lost. A 50% threshold was adopted: a customer is counted as lost when the estimated cancellation probability reaches or exceeds 50%. This value is not arbitrary — it corresponds to the minimum-error decision point (the Bayes threshold) for a binary outcome under symmetric costs: by classifying each customer by the individually more likely outcome, the expected number of misclassifications is minimized. Lower thresholds would be excessively conservative, since they would count as lost customers whose retention probability still exceeds their exit probability; higher thresholds would be excessively optimistic. An even more rigorous alternative would be the expected-value formulation, which weights each customer's profit by the retention probability instead of applying a hard cut; the 50% threshold was adopted for consistency with the operational convention of the notebook. In any case, the same threshold is applied to all models, so the comparison between them remains fair.

When these suggested premiums are applied as if they were readjustments on the current portfolio, every model generates relevant churn losses. Table 7 and Figure 3 show that Kernel Ridge produced fewer customers above the 50% churn threshold than Random Forest, but this comparison belongs to the current-portfolio scenario, not to the new-customer problem.

Model	Customers with churn $\geq 50\%$	% of base	Profit loss
K-Means (Clustering)	148	55.2%	R\$ 227,294.00
Linear Regression	140	52.2%	R\$ 229,115.84
Kernel Ridge	121	45.1%	R\$ 197,987.34
Random Forest (AI)	133	49.6%	R\$ 201,502.68

Table 7 - Simulated churn when applying model-derived prices to the current portfolio.

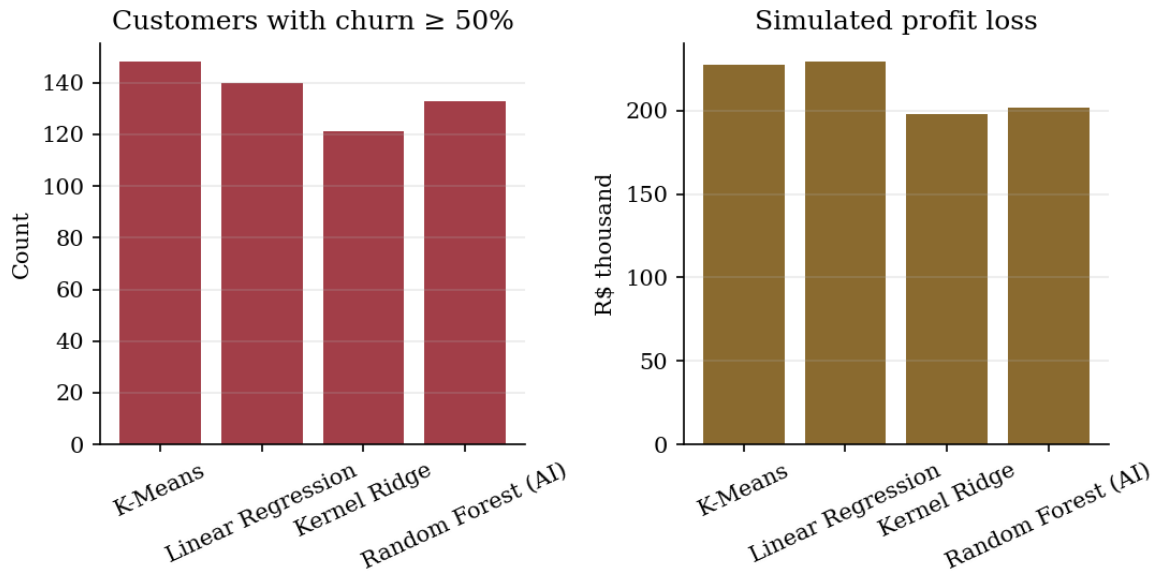


Figure 3 - Churn and profit loss by model in the readjustment simulation.

This result reinforces the article's thesis: a better expense prediction is not automatically the better readjustment policy. For current customers, the company already knows the current premium and the realized expense; the operational problem becomes deciding how much to readjust without exceeding a churn tolerance.

4.7 A specific policy for current customers

The current portfolio requires distinct treatment. Applying the predictive models' prices directly to existing customers would reduce profit below the current level, as shown in Section 4.6. The reason is conceptual: for a customer already in the base, the company does not need to estimate the expense — it already observes it —, nor does it need to estimate the premium, since it already charges it. The problem ceases to be prediction and becomes a readjustment decision under a churn constraint. For this scenario, therefore, a model-free approach is adopted.

The procedure has three steps. First, an acceptable churn target is fixed. Second, the Souza (2025) churn function is inverted to find which percentage readjustment produces exactly that churn, translating it into a new amount to charge each customer. Third, two safeguards are applied: never reduce a customer's premium below the amount already charged; and never charge below a floor of 9.52% margin over the expense ($\text{expenses} \times 1.0952$). Structurally loss-generating customers, for whom even this floor raises churn above 50%, are treated as lost — keeping a customer who generates a loss destroys value faster than losing them.

To make this procedure concrete, Table 8 applies it at the recommended target of 15% churn, detailing each step customer by customer. The first columns describe each customer's current situation — expense, premium charged, margin and profit — and the following columns show the amount to charge obtained by inverting the churn function, the implied percentage increase and the profit considering the resulting churn; six example rows are shown and the last one aggregates all 268 holdout customers. With the 15% target, the implied uniform readjustment is only 10.00% and churn-adjusted profit rises from R\$ 69,695.36 to R\$ 153,784.46 — a gain of R\$ 84,089.10, with each customer's churn chance kept at 15%.

Customer	Expense (R\$)	Current premium (R\$)	Current margin (%)	Current profit (R\$)	Amount to charge (R\$)	Increase (%)	Churn chance (%)	Churn-adj. profit (R\$)
1	R\$ 1,653.18	R\$ 1,944.92	15.0%	R\$ 291.74	R\$ 2,139.41	10.0%	15.0%	R\$ 486.23
2	R\$ 5,589.00	R\$ 5,322.85	-5.0%	R\$ -266.15	R\$ 5,855.14	10.0%	15.0%	R\$ 266.14
3	R\$ 5,159.52	R\$ 4,913.83	-5.0%	R\$ -245.69	R\$ 5,405.21	10.0%	15.0%	R\$ 245.69
4	R\$ 3,315.35	R\$ 4,420.46	25.0%	R\$ 1,105.11	R\$ 4,862.51	10.0%	15.0%	R\$ 1,547.16
5	R\$ 1,413.54	R\$ 1,662.99	15.0%	R\$ 249.45	R\$ 1,829.29	10.0%	15.0%	R\$ 415.75
6	R\$ 2,569.80	R\$ 3,426.41	25.0%	R\$ 856.61	R\$ 3,769.05	10.0%	15.0%	R\$ 1,199.25
Total	R\$ 771,195.59	R\$ 840,890.95	8.3%	R\$ 69,695.36	R\$ 924,980.05	—	—	R\$ 153,784.46

Table 8 - Per-customer detail of the readjustment policy at the recommended 15% churn target, for a sample of 6 of the 268 holdout customers; the last row aggregates the whole base.

Table 9 repeats the exercise at the theoretical ceiling of 49.9% churn. Each customer is charged considerably more — an implied increase of 23.96% — and the aggregate profit rises to R\$ 271,172.83, but each customer's churn chance comes close to 50%, a situation in which any calibration error pushes them into cancellation. The contrast between the two tables makes the central trade-off of the portfolio explicit: accepting more churn earns more, up to the point where the base starts to fall apart.

Customer	Expense (R\$)	Current premium (R\$)	Current margin (%)	Current profit (R\$)	Amount to charge (R\$)	Increase (%)	Churn chance (%)	Churn-adj. profit (R\$)
1	R\$ 1,653.18	R\$ 1,944.92	15.0%	R\$ 291.74	R\$ 2,410.92	24.0%	49.9%	R\$ 757.74
2	R\$ 5,589.00	R\$ 5,322.85	-5.0%	R\$ -266.15	R\$ 6,598.20	24.0%	49.9%	R\$ 1,009.20
3	R\$ 5,159.52	R\$ 4,913.83	-5.0%	R\$ -245.69	R\$ 6,091.18	24.0%	49.9%	R\$ 931.66
4	R\$ 3,315.35	R\$ 4,420.46	25.0%	R\$ 1,105.11	R\$ 5,479.60	24.0%	49.9%	R\$ 2,164.25
5	R\$ 1,413.54	R\$ 1,662.99	15.0%	R\$ 249.45	R\$ 2,061.44	24.0%	49.9%	R\$ 647.90
6	R\$ 2,569.80	R\$ 3,426.41	25.0%	R\$ 856.61	R\$ 4,247.38	24.0%	49.9%	R\$ 1,677.58
Total	R\$ 771,195.59	R\$ 840,890.95	8.3%	R\$ 69,695.36	R\$ 1,042,368.42	—	—	R\$ 271,172.83

Table 9 - Per-customer detail at the theoretical ceiling of 49.9% churn, for the same sample of 6 customers; the last row aggregates the whole base.

Tables 8 and 9 are only two points of a continuous curve. Figure 4 sweeps every possible churn target: churn-adjusted profit grows as more cancellation is accepted — charge more, earn more — up to the ceiling of 49.9%, which would raise profit from R\$ 69,695.36 to R\$ 271,172.83. The curve shows a plateau between roughly 24.9% and 40% churn: because of the regulatory discontinuity of the churn function at a +20% readjustment, these intermediate levels are unreachable and profit does not vary in that range.

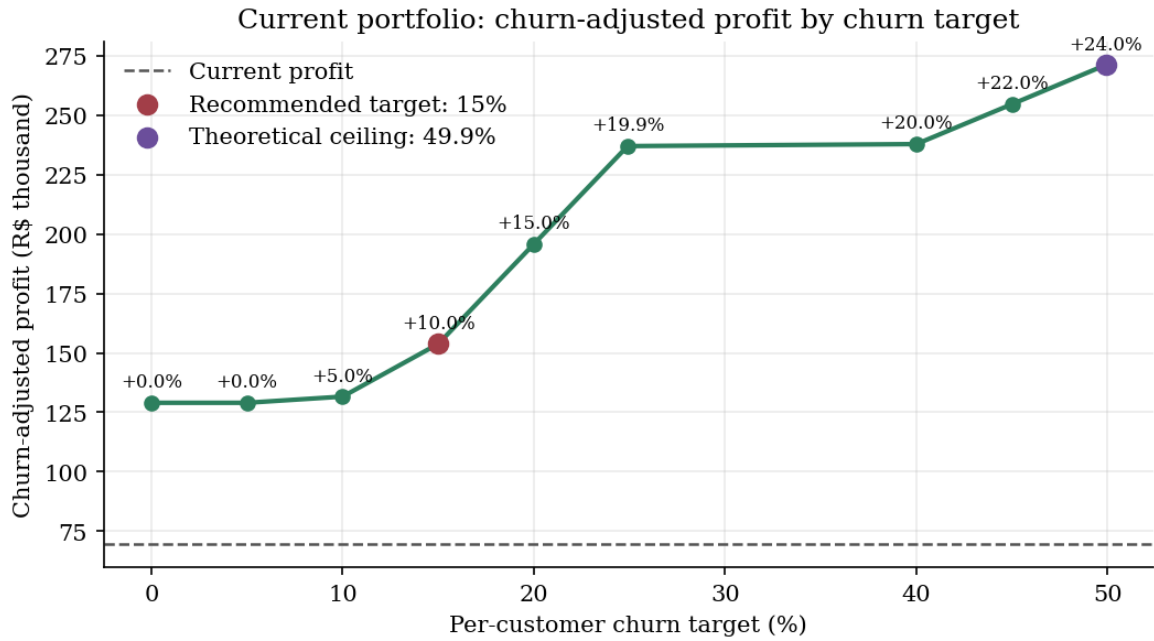


Figure 4 - Churn-adjusted profit for the current portfolio.

The 49.9% ceiling, however, is a limit, not a goal: it depends on the assumption that the churn calibration is exact for every customer and leaves no managerial safety margin. The recommended target is therefore 15% churn, from Table 8, a value anchored in the literature. Souza (2025) identifies a regulatory discontinuity at a +20% readjustment — the regulator's threshold that triggers the beneficiary's right to switch operators —, so keeping churn in the safe linear region requires readjustments well below that point. This target is further corroborated by private health-insurance market benchmarks, which point to annual retention around 83%, equivalent to 15-17% churn. The 10.00% readjustment of Table 8 stays comfortably below the discontinuity, preserving margin for calibration error and for future readjustments.

With the per-customer detail of Tables 8 and 9 and the risk curve of Figure 4 in hand, Table 10 summarizes the two decision scenarios — the recommended 15% target and the theoretical 49.9% ceiling — against the current situation of the portfolio.

Customer's churn chance	Implied increase	Churn $\geq$ 50%	Profit considering churn chances	Gain vs. current
Current	-	-	R\$ 69,695.36	-
Recommended target 15%	10.00%	0	R\$ 153,784.46	R\$ 84,089.10
Theoretical ceiling 49.9%	23.96%	0	R\$ 271,172.83	R\$ 201,477.47

Table 10 - Readjustment policy for current customers based on a churn target.

5 Discussion

The study shows that the evaluation of pricing models must separate three layers. The first is statistical: which model best predicts expenses on the holdout? In this layer, the AI-generated Random Forest was superior. The second is methodological: which model is most defensible within an explicit protocol of structural choice? In this layer, Kernel Ridge represents the best formal alternative found before the AI benchmark. The third is managerial: which pricing policy maximizes value without destroying retention? In this layer, the current portfolio requires its own churn rule.

The main caveat for managers is that a model's success on new customers does not authorize its direct application to current customers. If a company commissions a study to predict future expenses, the analyst adds value by also showing what to do with the existing portfolio. Leaving current customers without adjustment may mean losing money; on the other hand, readjusting everyone based on predicted expense may trigger churn. The solution is to treat the current portfolio as an optimization problem under a cancellation constraint.

The result also illustrates a realistic role for generative AI in data science. It can accelerate exploration, model selection and tuning. Even so, the professional value of the data scientist remains in validation, in correctly defining the problem, in critically assessing the metrics, in translating the model into an economic decision, and in governing its use.

6 Conclusion

This article presented a comprehensive mathematical modeling for health insurance pricing recommendation, guided by System Identification principles and benchmarked against an AI-generated workflow. The study showed that, for new customers, the AI-generated Random Forest achieved the best accuracy on the holdout. It also showed that Kernel Ridge was the best formal alternative before that benchmark, and that K-Means surpassing Linear Regression is a warning about linear misspecification in skewed and nonlinear datasets.

The practical conclusion is that the company should operate on two tracks: use the best validated model to price new customers, and use a churn-controlled readjustment rule for current customers. The scientific contribution lies in the framework, not merely in the algorithm ranking: prediction, price, churn and profit must be evaluated jointly.

7 Limitations

The main limitations are: the use of a public dataset of moderate size; the absence of detailed clinical variables; a churn function parameterized for simulation and not estimated directly on this dataset; the absence of temporal validation; and the comparison with a single AI workflow. In addition, the study does not replace actuarial, regulatory and ethical validation for real use by insurers.

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